



Name: Siddhesh ~~Sarode~~ Sarode
Roll Number: 09
Batch: 1
Experiment performed on (date): 11-09-2026

Title of the Experiment: Planck's Constant

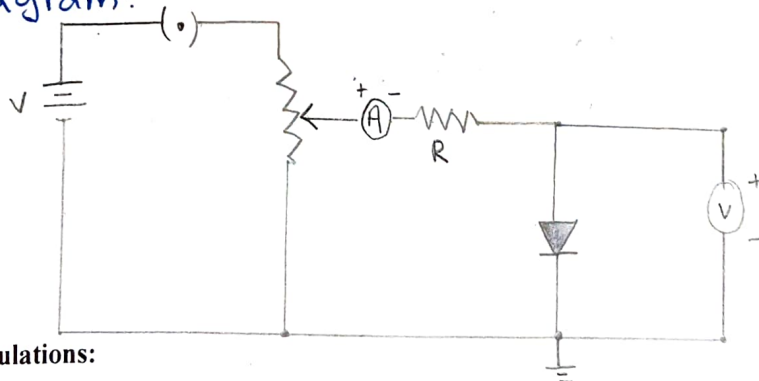
Aim: To determine the value of Planck's Constant using LEDs of different colors.

Apparatus: Power supply, rheostat, milliammeter, voltmeter, 1k resistor, various LEDs.

Observations:

Sr. No.	LED	Wavelength λ (nm)	$\frac{1}{\lambda}$ (nm ⁻¹)	V_{th} (Volt)
1	RED	650	1.538×10^{-3}	1.908V
2	YELLOW	570	1.754×10^{-3}	2.178V
3	GREEN	510	1.960×10^{-3}	2.434V
4	BLUE	469	2.132×10^{-3}	2.615V
5	IR	1096	0.912×10^{-3}	1.121V

Diagram:



Calculations:

$$\text{Slope} = \frac{y_2 - y_1}{x_2 - x_1}$$

$$= \frac{2.17 - 1.908}{1.75 - 1.538}$$

$$= 1.23$$

$$h = \text{slope} \times e$$

$$= \frac{1.23 \times 1.6 \times 10^{-19}}{3 \times 10^8}$$

$$= 6.56 \times 10^{-24} \text{ J/s}$$

$$\lambda = \frac{hc}{e} = \frac{hc}{eV_{th}}$$

$$\lambda_{IR} = \frac{hc}{eV_{th}}$$

$$= \frac{6.54 \times 10^{-34} \times 3 \times 10^8}{1.6 \times 10^{-19} \times 1.121}$$

$$= 1096 \text{ nm}$$

$$\lambda_{Blue} = \frac{6.54 \times 10^{-34} \times 3 \times 10^8}{1.6 \times 10^{-19} \times 2.615}$$

$$= 469 \text{ nm}$$

Result/s and Conclusion/s:

Value of Planck's constant (h) = $6.54 \times 10^{-34} \text{ Js}$

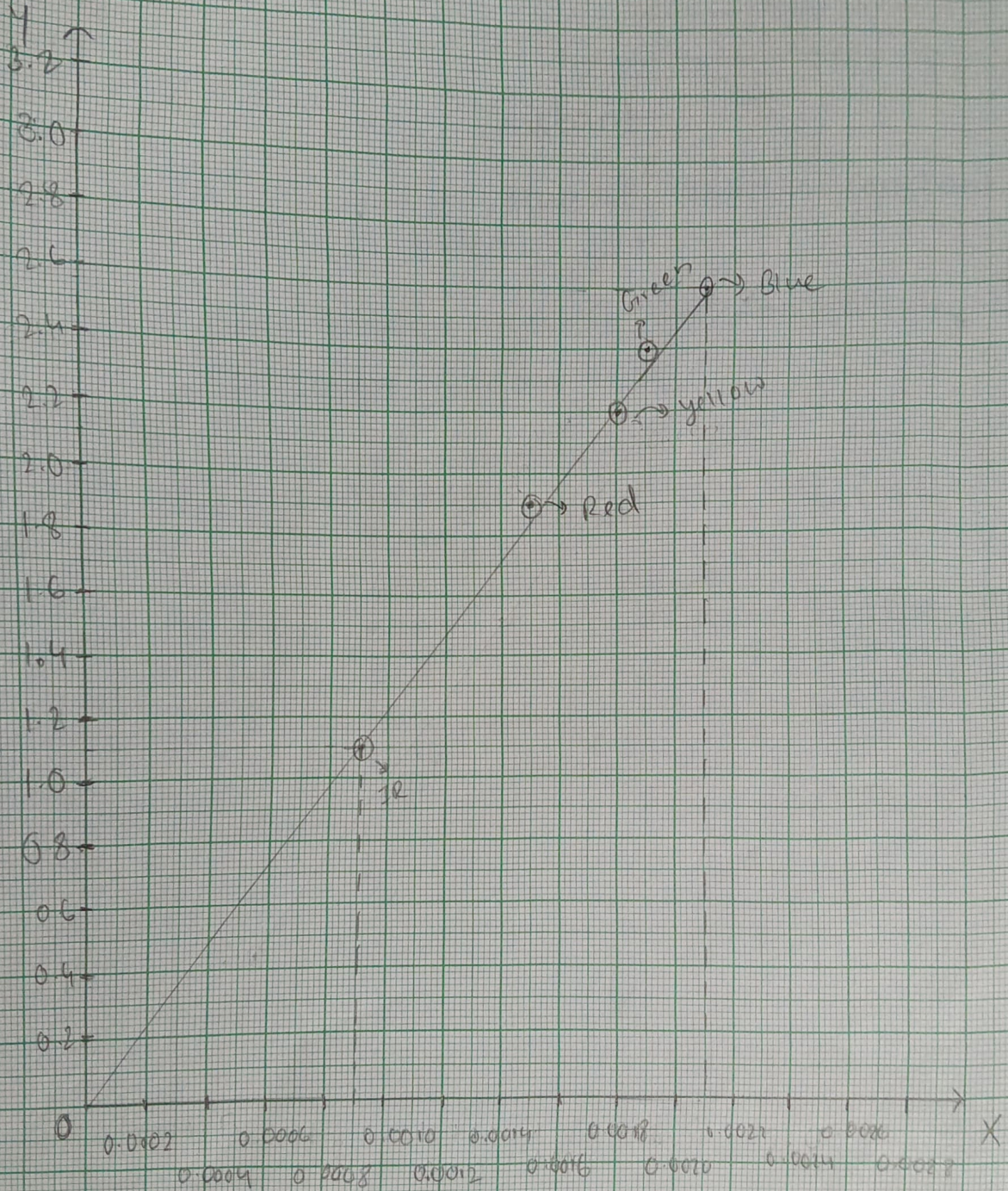
Assignment: Measure the threshold voltage for IR and Blue LEDs.

~~$$V_0 = \frac{hc}{\lambda}$$~~

Wavelength of IR LED = 1100 nm

Wavelength of Blue LED = 469 nm

V_{th} VOLT $\rightarrow$



$1/\lambda \rightarrow$
($\times 10^{-9}$)